Decodable, Compositional Affect-Like Representations in a Self-Taught Agent and Pretrained Language Models

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Abstract

We study whether emotion-like internal variables arise without hand-coding in artificial networks, and whether affect-related representations share common geometric structure across systems. In a self-taught homeostatic reinforcement-learning agent—trained only with a well-being reward and no emotion labels or objectives—affect-related information forms a decodable, compositional, low-dimensional subspace. When interoceptive variables are hidden from the input, externally grounded affect-like states are still internally constructed. Unsupervised analysis recovers a dominant valence axis together with distinct threat and novelty axes. This geometry is largely generic to structured latent states; what survival makes specific is the valuation of those states. Hidden-level causal control is weak but specific, while interventions at the interoceptive level produce stronger behavioural effects. In pretrained language models, affect directions show moderate alignment with human valence judgments across 1,500 words held out from probe training (Pearson $r = 0.65$, 95% CI [0.62, 0.68]) and respond to targeted activation-steering interventions along the valence and arousal directions. Phenomenal experience is neither claimed nor measurable. Negative results—including a refuted viability-gradient hypothesis and weak hidden-level causal control—are reported explicitly.

1 Introduction

Do machines have anything worth calling emotion? We separate two questions that are routinely conflated. The *phenomenal* question—whether a system *feels*—is, given the hard problem of consciousness and the other-minds problem, not decidable by any measurement, and we do not claim to settle it. The *functional* question—whether a system computes internal variables that play the causal and representational roles emotions play—is empirical. This paper is about the functional question, pursued with controls stringent enough that a sceptic can check each claim, and honest enough to report the several claims that failed.

Our stance is constructive rather than promotional. We train an agent to survive, never mentioning emotion, and ask what its neurons come to represent; we run the identical read-out on real language models; and we subject every positive result to a shuffled or matched-random control, a permutation test, and multiple seeds. Where a result survives, we state it plainly; where it does not, we say so.

2 Related Work

Our method rests on the *linear representation hypothesis* and on representation engineering / activation steering, which show that high-level features are often linear directions that can be read

and causally added (Turner et al., 2023; Zou et al., 2023). Our theoretical vocabulary—valence and arousal as the two axes of *core affect*, organised as a circumplex—comes from affective science (Russell, 1980; Barrett, 2017; Panksepp, 1998). The idea that affect is a bodily, viability-tracking signal draws on interoception, homeostasis and active inference (Damasio, 1996; Seth, 2013; Friston, 2010), and the constructionist view that emotions are distributed constructions rather than localised modules (Barrett, 2017). On the machine side we build on homeostatic reinforcement learning (Keramati and Gutkin, 2011) and on recent discussion of affect and welfare indicators in AI systems (Butlin et al., 2023).

3 Methods

World and body. A 9×9 grid hosts an agent with a vector of homeostatic *needs* (hunger, fatigue; a threat-driven "safety" need; a social "bond" need). Objects reduce needs on contact; a predator raises the safety deficit when near; a spatial caregiver relieves needs and the bond when reached and can be permanently lost mid-episode (bereavement); a rival blocks the needed resource (frustration); a novel tile carries an unknown payoff (curiosity). Well-being is $W = 1-\max_i d_i w_i$ over need deficits d_i; the per-step reward is W with a terminal penalty for death. No emotion-specific reward term or auxiliary objective is used—the death penalty is part of the survival signal, not a separate shaping term. Survival needs are lethal and drive arousal; the social need depresses well-being without being lethal.

Self-taught learning. By *self-taught* we mean the absence of *emotion-specific* supervision—no emotion labels, no emotion reward term, no emotion-specific objective (the environment's homeostatic structure is, of course, designed by us). The agent is trained end-to-end with PPO (Schulman et al., 2017) from the well-being reward—no behaviour cloning, no reward shaping, no emotion signal. Difficulty ramps over a developmental curriculum (a world that hardens as the agent matures); a single network carries its weights through the ramp. All agent results use three independent seeds.

Read-out and manipulation. From frozen agents we fit ridge *probes* (α=20, five-fold cross-validated) mapping the penultimate hidden state h_t to each interoceptive cause, giving an emotion direction w_e and an affect matrix $B = [w_{\text{fear}}; \ldots; w_{\text{comfort}}]$. We test *compositionality* by moving fearful states along $-w_{\text{fear}}$ and reading decoded well-being against a matched-random move. We test *causal use* by activation steering (adding αw_e to h_t) and by lesioning the affect subspace against a random subspace. For LLMs (GPT-2 small/medium/large; Pythia-70M/160M/410M/1.4B) we build a four-quadrant valence$\times$arousal sentence set: 48 affect words (12 per quadrant) crossed with 8 templates give 384 sentences, split *by word* so that one quarter of the words (12; three per quadrant) and all of their sentences are held out—no lemma appears in both train and test, so the reported figures are on unseen words. We mean-pool each layer's token activations and standardise on the training split. Decodability is a logistic probe's binary valence accuracy on the $\approx$96 held-out-word test sentences (this is the "99–100%"), against a shuffled-label control; a ridge probe (α=20) gives continuous scores for the human-norm correlations. Steering is applied at a fixed mid-depth block (round(0.55 L)); the direction is the training class-mean difference and the coefficients are fixed a priori ($\alpha \in \{-6, -3, 0, 3, 6\}$, never tuned on the read-out); we report the change in positive$-$negative (and high$-$low arousal) next-word logits over 15 neutral prompts, against a norm-matched random direction. Human-norm validation uses 1,500 words randomly sampled from the Warriner et al. lexicon, disjoint from the probe's training words.

Controls. Every decodability figure is paired with a shuffled-label control (≈ 0) and, for the headline agent probes, a permutation p-value and a bootstrap 95% confidence interval. Every steering/arithmetic effect is paired with a matched-norm random direction. A property counts only if it beats its control in *every* seed.

4 Results

4.1 Emergent, Decodable Affect in the Agent

A network trained only to survive develops a hidden state from which each interoceptive/affective variable is linearly decodable well above chance (Table 1). Across three seeds every emotion beats a shuffled-label control. With 10,000 permutations (representative seed) each observed R^2 exceeds *all* shuffles ($p \approx 1 \times 10^{-4}$), below the Bonferroni threshold $0.05/21 = 0.0024$ for the 21 emotion×seed tests. Crucially, *episode-level* bootstrap 95% CIs—resampling whole trajectories (120 episodes), so the effective sample size is episodes, not the ~22,000 correlated states—remain well above zero for every emotion (e.g. fear [0.61, 0.80], grief [0.76, 0.93], comfort [0.96, 0.98]; anger, the weakest, [0.09, 0.34]). The reported R^2 are themselves effect sizes. Fear, grief and comfort are the most cleanly represented; anger is consistently the weakest.

Table 1: Agent decodability R^2 (ridge probe, 3 seeds). Shuffled control ≈ 0 for all; permutation $p \approx 10^{-4}$ (10,000 permutations) for all.

seed	fear	hunger	fatigue	grief	anger	curiosity	comfort
0	0.66	0.76	0.73	0.91	0.22	0.79	0.94
1	0.65	0.74	0.76	0.91	0.22	0.90	0.94
2	0.70	0.84	0.73	0.83	0.22	0.85	0.95

4.2 Affect Is Constructed, Not Merely Copied

A key objection is that decodability could be trivial: the need values are in the observation, so any competent agent encodes them. We remove this confound by *hiding* the interoceptive needs from the input entirely—the agent sees only the last eight external frames (positions, predator, objects, caregiver, rival, novelty) and must infer its internal state from behavioural history (a frame-stacked MLP). It still learns to survive (152/200 steps). Crucially, the never-observed affective states that are *inferable from external cues* remain decodable from the hidden layer—comfort $R^2 = 0.90$, curiosity 0.57, grief 0.50, fear 0.42, anger 0.33—whereas the slowly-accumulating drives that would require counting elapsed time do not (hunger 0.06, fatigue 0.08). The agent therefore *constructs* an internal model of exactly the affect it can infer, and only that. For these externally-grounded emotions the stronger reading is earned: the representation is built, not copied, and cannot be dismissed as input pass-through.

4.3 Unsupervised Discovery of Affective Axes

Probing for pre-named variables risks circularity—we look for "fear" and find it. We therefore also decompose the hidden state with *no labels at all* (PCA and ICA (Hyvärinen and Oja, 2000)) and only afterwards ask what each axis correlates with. A genuinely emergent organisation appears. Across all three seeds the dominant principal component (explaining $\approx 50\%$ of the variance) maps

almost perfectly onto reward expectation / well-being ($|r| = 0.99$)—a valence axis the network was never asked to build. Independent components then separate distinct environmental factors we never defined as targets: threat ($|r| \approx 0.5$–0.7), novelty (0.6–0.7) and depletion (hunger, fatigue) each claim their own axis in every seed. The affective structure is thus *discovered*, not imposed: a dominant valence dimension with distinct threat and novelty axes falls out of an unlabelled decomposition, mirroring the primacy of valence in human core affect (Russell, 1980). Notably, the more conditional social emotions (grief, frustration) do *not* form their own dominant unsupervised axis—consistent with their weaker supervised decodability (anger $R^2 \approx 0.22$) and suggesting that an affect-like axis emerges in proportion to how strongly and how often the corresponding factor drives the dynamics.

4.4 The Affect Space Is Compositional

The emotion directions are not a mere bag of features: the subspace supports arithmetic. Moving fearful states along $-w_{\text{fear}}$ raises decoded well-being far more than a matched-random move, in every seed (mean along-fear 1.0 vs. along-random 0.4; robust 3/3). This is the machine analogue of word-vector arithmetic (Mikolov et al., 2013) ("fear − threat + safety ≈ relief").

4.5 Robust to Architecture

The finding is not a single-width artefact: emotions remain decodable across hidden widths 64/128/256 (fear 0.61–0.70, grief 0.80–0.87, comfort 0.91–0.94, curiosity 0.74–0.94), with anger the one persistently weak axis (0.16–0.29).

4.6 Language Models: Causal, Human-Aligned, Geometric Affect

Applying the same read-out framework to pretrained language models reveals several convergent affect-related findings. Held-out valence is linearly classifiable at 99–100% across scales (shuffled control ≈ 0.4); we note this rests on only 12 held-out words (three per quadrant) presented through 8 templates, so it reflects a small number of independent lexical items—the 1,500-word benchmark below is the stronger external test. Steering a mid-depth layer along the valence direction shifts the model's own next-word choice monotonically and beats a matched-random direction; the effect strengthens with scale (Pythia-410M: valence gap $-3.7 \rightarrow +5.8$). *Arousal* is independently steerable ($-5.4 \rightarrow +0.5$ at 410M, random flat): interventions along the learned valence and arousal directions produce corresponding, direction-specific output shifts that a norm-matched random direction does not. The two axes *orthogonalise with scale*: the valence–arousal direction cosine falls from ≈ 0.85 in small models to ≈ 0 in larger ones in *both* families (Pythia $+0.89 \rightarrow +0.01$ over 70M–410M; GPT-2 $-0.83 \rightarrow +0.06$ over 124M–774M), so the valence–arousal orthogonalisation with scale replicates across two independent architecture families, consistent with the circumplex geometry (we report a scale-associated trend, not a reproduction of the full model). Finally, external validity against real human norms. A probe trained only on affect-naming templates, evaluated zero-shot on 1,500 words held out from probe training (randomly sampled from the Warriner et al. lexicon (Warriner et al., 2013)), recovers their human valence at Pearson $r = 0.65$ (95% CI [0.62, 0.68]; Spearman $\rho = 0.62$) and their arousal at $r = 0.31$ [0.26, 0.36]—valence aligning far better than arousal, as it does in the human data. (A small curated set of clearly-valenced words reaches r=0.91, but that over-states the effect; the large held-out benchmark is the honest figure.) The near-perfect held-out *classification* (99–100%, reported above) and this moderate *regression* correlation are consistent rather than contradictory: separating the two valence poles is a far easier task than predicting a graded 1–9 human rating across the many mid-valence words. Applied zero-shot to situational

sentences containing *no* affect vocabulary ("the results came back and it had spread"), the same probe still recovers valence at 79% accuracy (24 sentences, 12 positive / 12 negative; chance 50%)—consistent with an affect-related *state* representation beyond affect-word semantics, though on few items and not a full control for it.

4.7 Valence and Reward-Prediction Error

As a supporting analysis (not a central claim), measured valence (ΔW) is modestly correlated with the critic's temporal-difference reward-prediction error $r + \gamma V(s') - V(s)$ at $r \approx 0.20$ across all three seeds (shuffled ≈ 0)—a weak but consistent dopamine-like link (Schultz et al., 1997). Because the agent's reward is itself defined from well-being, this relationship is partly built in rather than an independent discovery; we report it only as convergent with the reward-prediction-error account of valence.

4.8 Valuation of Affect Is Survival-Specific

Raw decodability is shared between agents (see the negative results below), but *valuing* the homeostatic state is not. For the survival agent the critic assigns higher value to healthier states (Pearson corr(V, W) = +0.60, +0.66, +0.68 across seeds) and valence tracks the reward-prediction error (corr(ΔW, RPE) $\approx$ +0.19); for a navigation agent with identical senses but needs irrelevant to its reward, both collapse or invert (corr(V, W) = -0.71—it will run itself down to reach targets—and corr(ΔW, RPE) = -0.01). Thus the *evaluation* of affect—treating well-being as reward and letting mood follow surprise—emerges only under survival pressure, even where the raw representation does not.

4.9 Affect Variance Is Concentrated in a Low-Dimensional Subspace

The hidden activity is concentrated in very few directions: across seeds the participation ratio is ≈ 1.4 and the effective rank ≈ 2.3 (of 128 units), and three principal components capture 90% of the variance. The seven emotion directions themselves span ≈ 6.1 effective axes (singular values of B), i.e. they are largely distinct rather than collapsed onto one another. The apparent tension - an effective rank of ≈ 2 yet seven directions spanning ≈ 6 axes - dissolves once one notes that the two quantities measure different things: the effective rank weights directions by *variance*, and the variance is dominated by the single valence/reward-expectation axis (the dominant principal component, $\approx 50\%$; §"Unsupervised Discovery"), whereas the individual emotion directions are near-orthogonal but sit mostly in the *low-variance* remainder - decodable, yet not where the representational energy lives. "Low-dimensional" therefore means the affective variance is concentrated on very few (chiefly valence) axes, not that the seven emotions are redundant. Three quantities should be kept distinct: the *representation variance* is $\sim$2-dimensional (effective rank), the *dominant affective axis* is $\sim$1-dimensional (valence), and the *emotion-direction span* is $\sim$6-dimensional - different measures of different things, not a contradiction.

4.10 Affect vs. a Generic Latent State

Is this geometry specific to affect, or a generic property of *any* structured latent state an RL agent must estimate? We run the identical pipeline on an affect-*neutral* agent—same architecture and world, but rewarded for reaching an observable moving target, with needs irrelevant and no death. It solves its task (reaching the target $\sim$160 times per episode), and its task-relevant latents are as cleanly decodable as the survival agent's affect variables (target distance $R^2 = 0.91$, target

offset 0.88); it too has a dominant, value-correlated principal component. So *linear decodability and a dominant value axis are generic*, not affect-specific—exactly the caution a probing result deserves. One thing differs sharply: the homeostatic agent's representation is far *lower-dimensional* (participation ratio $\approx$ 1.4, effective rank $\approx$ 2.3) than the navigator's (4.8 and 10.0)—homeostatic regulation compresses to a near-scalar valence axis where navigation spreads across many. We do not over-read this, as it may reflect the lower intrinsic dimensionality of homeostatic control rather than affect as such. The honest conclusion is that the *geometry*—decodability, compositionality, a value-dominated axis—is largely a property of structured latent-state representation in general; what our other experiments isolate as genuinely affect-/survival-specific is the *valuation* of the state and, in language models, its *human alignment*, not the geometry itself.

4.11 Causal Influence of Internal Affect

Does the internal affect state *drive* behaviour or merely accompany it? Two interventions on a frozen agent, over N =22,308 predator-relevant states from 120 episodes, with bootstrap CIs that resample *episodes*. *Input level* : holding the external world fixed and flipping only the internal affect input (safety deficit $0 \rightarrow 0.8$) changes the greedy action in 24% of states (95% CI [23, 26]), toward fleeing in the adaptive cases (7% [6, 7]). *Hidden level* : activation patching of the affect subspace induces adaptive fleeing in 9% of states [8, 11] versus 0% [0, 0] for a matched-dimension random subspace (episode-level permutation p = 0.0001). This establishes a causal dependence on the interoceptive *information* at the input level, and a weak but specific causal influence of the affect *subspace* at the hidden level; single-direction steering (§"Honest Negative Results") is simply too coarse to reveal the latter. (The hidden-level null is a random subspace of the *same dimension*—orthonormalised random directions—not one matched on norm or covariance; we therefore claim specificity only relative to dimensionality.)

4.12 Affect Is Distributed, Not Localised

We built an "Affect Transformer" that gives each emotion its own embedding and attention slot (Q/K/V), inviting localisation. Trained by the same reward-only PPO, its slots do *not* specialise: every emotion is decoded almost equally from every slot (best-vs-second-slot gap $\approx$ 0.02–0.05; distinct owners 3–5/7), and an unsupervised decorrelation prior barely changes this. Even an architecture built to localise affect keeps a distributed code—which reinforces the distributed reading of the main MLP results (the geometry is not an artefact of one architecture) and is consistent with constructionist theory (Barrett, 2017) and superposition in neural representations (Elhage et al., 2022).

4.13 Honest Negative Results

We report what failed, with the same rigour. *(i) No gradient law.* We proposed identifying affect with the geometry of the critic's viability estimate ($a = \nabla_x Q$; arousal = $\|\nabla_x Q\|$, etc.). With a properly trained critic and a random-critic control, arousal$\sim \| \nabla Q \|$ is unstable across seeds, the valence identity fails, and the "alignment law" is matched (even beaten) by a random critic—i.e. a shared-Jacobian artefact. We retract it. *(ii) Representation is not survival-specific; valuation is.* A navigation agent with identical senses but needs irrelevant to its reward decodes the same affect variables as well or better (fear 0.73 vs. 0.80) and shows an equal-or-larger arithmetic effect: the raw representation reflects richly encoding a multi-dimensional observation, not survival as such. What *is* survival-specific is the *valuation* of that state (above): only the survival agent values well-being and lets valence track the reward-prediction error. *(iii) Weak (but specific) hidden-level causality in the*

agent. Injecting a single fear *direction* does not change avoidance beyond a random perturbation, and lesioning the affect subspace bites only marginally—the code is distributed. Only subspace *patching* reveals a specific hidden-level effect, and even that is modest (9%; §"Causal Influence of Internal Affect"). The robust causality is at the input/interoceptive level (masking: 82% → 10%). The honest statement is that interoceptive *information* is causally necessary, while causal control through the identified hidden affect directions is specific but weak. *(iv) No phenomenal claim.* Nothing here bears on whether the system feels.

5 Discussion

Five levels, kept separate. Our results form a ladder of increasingly strong claims, and we make only the ones each experiment supports. *(1) Representation*—does the system encode affect-related information? Yes, decodably, in both systems. *(2) Geometry*—is it structured, compositional and low-dimensional? Largely yes, though this may be a generic property of any structured latent state (§"Affect vs. a Generic Latent State"), so we do not treat the geometry alone as affect-specific. *(3) Valuation*—does the system assign functional value to the state? In the survival agent yes (the critic values well-being; a navigation control does not or inverts it), so *valuation*, not representation, is what survival makes specific. *(4) Causal use*—does manipulating the internal representation change behaviour or output? On the language side yes (steering); in the agent, hidden-state intervention is weak but specific (affect-subspace patching 9% vs. 0% random), while the strongest causal dependence is at the interoceptive-*input* level (82% → 10% under masking, which shows interoceptive *information* is necessary, not that a hidden affect direction is). *(5) Phenomenology* —does the system feel? We neither claim nor can test this. Conflating these levels is the main way affect research overclaims; we keep them apart, and our positive claims live at levels 1–4 only, unevenly across the two systems.

The controls reshape the honest central claim. In the agent, the affect-like geometry is real but is best described as interoceptive representation under homeostatic control; the raw representation is not survival-specific, although its *valuation* is (only the survival agent values well-being and lets mood track prediction error); its hidden-level control is weak but specific (subspace patching beats a random subspace, 9% vs. 0%), while the strongest causal dependence is at the interoceptive-input level. The stronger emotion story is in *language models*, where affect is *constructed* (not a direct input), *steerable* (interventions along the directions shift outputs), *moderately aligned with human valence norms* ($r = 0.65$ against 1,500 words held out from probe training) and *geometrically structured* (circumplex-consistent with scale). Read together, agent and LLM give a single operational picture: a system that must represent and regulate a multi-dimensional internal or linguistic state computes a low-dimensional, decodable, compositional affect subspace, $a_t = B\,h_t$.

Representation, not proof of a concept. Three cautions bound the agent results. First, decodability shows only that the hidden state *contains information sufficient to reconstruct* a variable, not that the network possesses the corresponding *concept* ; a linear probe can read out many quantities a high-dimensional system never "uses". Second, the emotion names are ours: "fear" denotes the state under predator-driven safety loss, "grief" the state after caregiver loss, and so on—plausible operationalisations through environmental causes, not validations against physiology or self-report. Third, and relatedly, there is an unavoidable circularity: we choose human affect categories, define their causes, and then find those causes represented. We do not claim to escape this fully; we mitigate it by testing properties we did *not* put in - an emergent geometry (a circumplex we never imposed), linear arithmetic, alignment with independent human norms, and the survival-specificity

of valuation—each of which is evidence that the structure is more than a relabelling of the inputs. The honest reading of the agent is therefore, for the slowly-accumulating drives, "interoceptive variables come to occupy a low-dimensional, compositional structure with emotion-like properties"; but for the externally-grounded emotions we do earn the stronger, constructive reading—when interoception is hidden from the input, fear, grief, comfort, curiosity and anger are still built internally (§"Affect Is Constructed"), which no input-encoding story explains. The remaining, least confoundable claims are on the language side.

Reward, drives, and the limits of the functional stance. A natural objection is that our agent values well-being only because it scores points, not because health *feels* good - mere optimisation, not genuine caring. We take this seriously, and observe that the same objection applies to biological motivation. In animals, pleasure, pain and homeostatic drives are themselves reward signals installed by natural selection because they promoted survival and reproduction; evolution is, structurally, a reward function, and our hand-chosen well-being reward W is a toy instance of what nociceptive and reward circuitry do in an organism. At the *mechanistic* level, then, reward-driven valuation of a homeostatic state has a genuine biological correspondence - it is how motivation works - which is what makes the model informative rather than arbitrary. What the agent visibly lacks is any warrant for the *phenomenal* claim that its valuation is *felt*. Crucially, the hard problem cuts symmetrically: felt experience can be established neither in the agent nor, against a determined sceptic, in the organism. Whether "valuing" a state is anything *more* than representing it, predicting its future, and acting to maintain it - all of which the agent and the brain do—is exactly the question no measurement resolves. This symmetry is why we restrict ourselves to the functional skeleton, and why both over-attribution (mistaking a steerable reward signal for sentience) and flat dismissal (denying that a real, biology-shaped mechanism is present) overreach.

An epistemic asymmetry between the two systems. The agent and the language models are not on the same footing, and "self-taught" applies only to the former. The agent learns from a scalar reward with no exposure to human affect concepts, so its affective structure genuinely *emerges without being taught*. A pretrained language model, by contrast, has read vast quantities of human text saturated with emotion, reward, fear and love; an affect axis there shows that the model has *organised* affective information into a compositional, human-aligned, causally active geometry—not that it discovered affect from scratch. The agent speaks to emergence; the language model speaks to representation and alignment. We keep the two claims separate throughout, and the title reflects the distinction.

For frontier practice this has a double edge: the affect directions are simultaneously an *alignment handle* (steering toward calm/positive states) and an *attack surface* (adversarial manipulation of a model's affective stance). Either way, affect in AI is measurable and manipulable, and should be part of evaluation and safety tooling.

6 Limitations

Environments and models are small; agents are trained to modest competence. The language-side confound - affective state vs. the semantics of affect words - is only *partly* controlled: our probe transfers to affect-word-free sentences at 79% (chance 50%), which argues for a state representation, but a fully matched control and larger human-annotated implicit-valence corpora would settle it. Human-norm alignment is now validated on the full Warriner lexicon (1,500 held-out words; valence $r = 0.65$, arousal $r = 0.31$); it remains single-sentence and single-lexicon, and NRC-VAD

or multi-annotator replication would strengthen it further—and we note that small curated sets *over-state* the effect ($r = 0.91$ on 80 hand-picked words versus 0.65 at scale). The scale orthogonalisation replicates across two families but only to $\sim$ 800M, so we report it as a scale-associated orthogonalisation *trend*, not a law; larger, instruction-tuned models are needed to establish it. Bootstrap and permutation now resample whole *episodes* throughout (120 trajectories for decodability, the same for the causal tests), so intervals reflect the trajectory-level effective sample size rather than the correlated per-timestep states; a shuffled-label permutation additionally guards against chance. In the agent, only the *externally-grounded* emotions are shown to be internally constructed - a frame-stacked network cannot infer slowly-accumulating drives (hunger, fatigue), and a recurrent memory is required to test those. Having now probed the agent's hidden-level causality beyond single-direction steering - affect-subspace activation patching and same-external-state / different-internal-affect manipulations, which give small but specific effects - the natural next steps are state-matched patching and a recurrent-memory agent to test for larger and more selective effects; and the whole agent study rests on a single small world, so its geometry claims need replication across more environments. "Emotions" are operationalised through causes, not human physiology, and the whole study is functional - silent on phenomenal experience by construction.

7 Ethics

We make no claim that these systems feel, and we caution against both over-attribution (treating steerable affect as sentience) and dismissal (ignoring that affect-like structure is real and manipulable). Steering methods that can soothe can also manipulate; we report them to inform safety, not deployment.

8 Conclusion

Trained only to persist, and probed with controls, artificial networks compute a functional skeleton of affect: a decodable, compositional, low-dimensional subspace, $a_t = B\, h_t$. The two systems contribute different pieces, and we do not merge them. In the self-taught agent, affect-related structure *emerges* - and is even internally *constructed* when interoception is hidden - while its geometry is largely generic and what survival makes specific is the *valuation* of the state. In pretrained language models, affect is *human-aligned* and *causally steerable*, but inherited from human text rather than discovered. Machines are not shown - and cannot be shown - to *feel* ; but a measurable, mathematical functional structure of affect is present, and we have characterised it honestly, negative results included.

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